

Database Requirements Document

Project: Online Examination System

1. System Overview

The purpose of this database is to provide a robust, scalable, and centralized storage solution for the Online Examination System. The database is designed to manage academic hierarchies, course distributions, examination life cycles, and student performance tracking. It ensures data integrity and consistency while supporting administrative reporting and real-time exam processing.

2. Functional Requirements

The database management system (DBMS) shall facilitate the following functional areas:

- **Student Management:** Track student profiles and their respective department affiliations.
- **Department Management:** Organize courses and students within specific academic departments.
- **Instructor Management:** Maintain records of faculty members and their instructional roles.
- **Course Administration:** Manage a catalog of courses, including their topics and curriculum mapping.
- **Examination Life Cycle:** Store various exam types, associated questions, and multiple-choice options.
- **Enrollment & Assignment:** Handle M:N relationships between students/courses and instructors/courses.
- **Performance Tracking:** Record student answers in real-time and compute final grades/results.

3. Data Entities

- **Student:** Stores personal details, academic ID, and department linkage.
- **Department:** Represents academic divisions within the institution.
- **Instructor:** Contains faculty profiles and professional details.
- **Course:** Stores individual course metadata (Title, Credit Hours).
- **CourseTopic:** Breaks down courses into specific modules or themes.
- **Exam:** Defines exam instances, durations, and total marks.
- **Question:** The central repository for exam items, categorized by type/difficulty.
- **Choice:** The set of possible answers for each question.

- **StudentAnswer:** Transactional record of selections made by students during an exam.
- **Grade:** Final calculated results for a student's attempt at an exam.
- **StudentCourse:** Intersection entity for student enrollments.
- **InstructorCourse:** Intersection entity for teaching assignments.

4. Relationships

4.1 One-to-Many (1:N) Relationships

- Department ↔ Student: A department hosts many students.
- Course ↔ CourseTopic: A single course consists of multiple topics.
- Exam ↔ Question: An exam is composed of several questions.
- Question ↔ Choice: Each question provides multiple options for selection.

4.2 Many-to-Many (M:N) Relationships

- Student ↔ Course: Facilitated via 'StudentCourse' (Enrollment table).
- Instructor ↔ Course: Facilitated via 'InstructorCourse' (Teaching assignment table).

5. Business Rules

- Departmental Integrity: Each student must be assigned to exactly one department.
- Course Enrollment: Students can enroll in multiple courses, but cannot have duplicate active enrollments for the same course instance.
- Teaching Load: Instructors can be assigned to multiple courses; a course can have multiple instructors.
- Exam Structure: Every exam must contain at least one question; every question must have at least two choices.
- Submission Rules: A student is permitted only one submitted answer per question per exam attempt.
- Unique Identification: All primary entities must have a system-generated unique identifier (Identity Column).

6. Database Design Notes

- Platform: Microsoft SQL Server
- Normalization: All tables are designed to meet 3rd Normal Form (3NF) to eliminate data redundancy.
- Primary Keys (PK): Every table uses a non-null Unique Identifier (INT IDENTITY).
- Foreign Keys (FK): Enforced Referential Integrity on all relational links with 'ON DELETE NO ACTION' to prevent accidental data loss.
- Composite Keys: Used in 'StudentCourse' (StudentID, CourseID) and 'InstructorCourse' (InstructorID, CourseID).

- Data Types: Use VARCHAR/NVARCHAR for strings, INT for IDs/counts, DATETIME for timestamps, and DECIMAL for grades.

7. Reporting Requirements (Stored Procedures)

1. sp_GetStudentsByDept: Filters student roster based on DepartmentID.
2. sp_GetStudentGrades: Retrieves all exam results and final percentages for a specific student in a specific course.
3. sp_GetInstructorLoad: Lists all courses taught by an instructor along with the current count of enrolled students.
4. sp_GetCourseCurriculum: Displays all topics and sub-topics associated with a specific CourseID.
5. sp_GetExamPaper: Fetches a complete set of questions and their respective choices for an ExamID.
6. sp_GetStudentExamDetails: Detailed audit of a student's selected answers vs. correct answers for post-exam review.